Sensitivity-kernel acceleration

Timings, validation and remaining limits on Tunya

The approved acceleration campaign is complete. The latest native candidate reduces the degree-1536 unprofiled runtime from 536.442 to 313.361 seconds (41.6%), with byte-identical scientific outputs. The previous accelerated candidate remains faster at degrees 128 and 384. Global synthesis/storage/threading checks pass internally, but its frozen comparison against the original synthesis route still fails. No sensitivity-kernel figures or requested high-degree complete 179-side global volume were generated.

1. Fixed experiment and comparison domains

The observable is windowed vertical-displacement energy,

$$E = \frac{\Delta t}{2} \sum_j H_{1000\text{–}1200}(t_j)\, u_z(t_j)^2.$$

The PREM500 model uses a source at 10°N, 20°E, depth 50 km, and receiver at 45°N, 75°E, depth 3 km. The nominal band is 0–10 mHz, flat to 8 mHz and tapered to 10 mHz. Sampling is 5 s, record duration 3600 s, FFT length 1024 and period 5120 s; damping is $5/3600\ \mathrm{s}^{-1}$. There are 52 physical bins including DC, plus one zero-weight guard bin in response capture. The SEM uses six GLL points per element and radial refinement 0.05, with 179 positive native element/node sides. Both sides of interfaces remain distinct. Contacts are off; source-shell contact contributions are omitted.

At high degree the native DSpec/YSpec comparison uses the same **64-angle, 179-side diagnostic**, not a complete global volume. Global diagnostics use the complete product-bandwidth grid, with $n_\theta = 2L + 1$ and $n_\phi = \mathrm{FastFFTSize}(4L + 1)$, on eight sides at low degree, two at $L = 128, 384$, and one fluid side at $L = 1536$. Global timings must not be compared to YSpec's 64-angle timing as if their output domains matched.[1]

2. Native timings and individually measured changes

L	Original (s)	Candidate v4 (s)	Candidate v5 (s)	v5 reduction
128	41.856	26.912	29.868	28.6%
384	123.951	80.847	86.122	30.5%
1536	532.984	370.683	314.663	41.0%

All entries above are instrumented process wall times. Degrees 128 and 384 use the isolated paired reruns for v4/v5; degree 1536 uses the accepted full component runs. At 1536 the unprofiled v4/v5 runs take 369.677/313.361 s and agree byte-for-byte with their instrumented builds. Native outputs also match the original production route through a verified hash chain, despite the matched-response control having a different baseline capture setting.

At 128 and 384, v5 is respectively 11.0% and 6.5% slower than v4 in the isolated repeats. The regression is concentrated in dynamic assembly, scaling and numerical LU, with identical work counters. Its underlying compiler/allocation/cache cause has not been established. The low-degree result cannot be attributed solely to earlier overlapping work.

The retained changes are shared coefficient-label layouts/permutations, nonradial response reuse, exact-pattern symbolic reuse with fresh numerical LU, and removal of two unused full-order zero adjoint matrices. The radial $L = 0$ weighted solve is retained to preserve the frozen numerical result. The separate native-node recovery prototype was rejected: at 1536 it took 414.657 s versus 400.374 s for packing alone, and increased the recovery component from 64.631 to 78.965 s.

[1] Timings are individual observed runs on the same Tunya host, not statistical estimates or comparisons at matched local-kernel accuracy. Inclusive profiler scopes overlap; only exclusive entries may be summed. All stages at every measured degree are retained in the accompanying JSON and CSV.

The last two-line allocation removal avoids about 2.15 TB of cumulative zero-fill traffic in the pole-only cached route at degree 1536. The source/solver exclusive remainders fall by 64.529 s; the observed net wall saving is 56.021 s. It is retained for its measured high-degree benefit, with its low-degree regression disclosed.

Exclusive component, $L = 1536$	Original (s)	v4 (s)	v5 (s)
Coefficient packing/reorder	134.740	2.607	2.716
Layout deduplication	0	0.548	0.548
Dynamic matrix assembly	31.828	16.116	16.276
Numerical LU	98.255	50.591	52.777
Symbolic analysis	23.257	2.79e-04	2.88e-04
Equilibration/scaling	23.019	11.223	11.769
Forward back-solves	27.815	14.158	14.122
Adjoint back-solves	8.955	9.196	9.071
Native strain/derivative recovery	64.612	64.962	66.655
Angular Eigen multiplication	64.685	63.747	63.930
Receiver basis/QR	5.581	11.880	11.706
Spheroidal source/solver remainder	12.757	75.805	21.302
Toroidal source/solver remainder	1.984	12.744	2.718
Final pointwise products	0.009	0.010	0.010

The historical recovery timer still contains “interpolation” in its name. At native SEM nodes, the sampling plan selects the node and applies cached D_{ref}/J derivative weights to the current modal displacement fields. It does not require arbitrary off-grid radial output. Gathering modal values, applying these weights, constructing strains and allocating small coefficient records remain real work. The rejected stack-allocation prototype did not implement a successful fully batched derivative/recovery algorithm.

At degree 1536, factorisations fall from 322665 to 162921; only 3 symbolic analyses are performed, with 162918 reuses. Nonradial response capture has 162816 misses and 159744 later hits; 105 radial bypass calls retain the original solve. Logical retained response storage is 10.901 GiB within the 32 GiB budget, and observed profiled-native peak RSS is 8.736 GiB. There is no cache fallback.

3. YSpec comparison at the same native diagnostic points

L	Original DSpec (s)	YSpec waveform (s)	YSpec kernel (s)
128	41.866	16.843	77.177
384	122.825	85.677	321.047
1536	536.442	1061.726	3338.354

These are the saved unprofiled profile-06 results. YSpec’s independent waveform run is separate; its kernel diagnostic uses the frozen shared DSpec measurement baseline. Their times are not added and presented as a measured end-to-end YSpec run. DSpec’s kernel timing includes its own baseline. The latest unprofiled DSpec degree-1536 result is 313.361 s, compared with the saved YSpec kernel diagnostic’s 3338.354 s, subject to these methodology and scientific limitations.

YSpec exclusive component, $L = 1536$	Seconds
Upward integration	1005.270
Downward integration	57.929
Spheroidal b-vector transport	2195.872
Matching/response transport	12.050
Coefficient/strain recovery	11.706
Source angular synthesis	37.409
Adjoint angular synthesis	22.449
Elastic contraction	0.081
Parameter/objective products	0.099

YSpec's cost is dominated by radial integration and b-vector transport, while the accelerated DSpec cost is shared between sparse solves, native strain recovery and angular synthesis. Final contractions are small in both native diagnostics. This is a comparison of the configured algorithms and tolerances, not proof of equal local-kernel accuracy.

4. Global synthesis: complete angular grids on selected sides

L	Wigner mode	Sides	Wall (s)	Peak RSS (GiB)
128	generated	2	31.968	0.190
128	stored	2	30.424	0.342
384	generated	2	104.868	0.744
384	stored	2	100.433	4.927
1536	generated	1	509.233	6.860
1536	stored	1	483.668	276.787

All runs request one outer worker and 32 inner GSH threads, with other numerical-library thread limits controlled. The maximum observed resident count is 34 including the supervisor. The source releases the main-thread OpenMP pool after grid construction and before the worker team is formed; runtime cleanup succeeds. The global cap is 40 and the waiting follow-up cap is 4, keeping the configured aggregate below 50. Supervision is userspace observation/enforcement, not kernel aggregate accounting.

Threading is harmful on the tiny cases: Generated degree-3/12 runs rise from 3.502/5.242 s serial to 21.939/27.510 s with 32 threads. Keep those cases serial. At degree 384, the serial Generated/Stored runs took 259.723/173.746 s; the 32-thread versions take 104.868/100.433 s. There is no completed serial degree-1536 reference: its earlier unfinished run was deliberately stopped and preserved at 1952.070 s. Do not infer a 32-fold speedup or turn that partial duration into a completed benchmark.

The degree-1536 Stored table is about 270.44 GiB; observed total peak RSS is 276.787 GiB, versus 6.860 GiB Generated. Stored saves only 25.565 s (5.0%) in this one-fluid-side end-to-end case. Generated is the sensible default for the initial bounded outputs; many-side cache amortisation remains a separate measured decision. Both modes spool 11.293 GiB of modal records for all native sides, while the one-side kernel output is 0.846 GiB.

Degree-1536 wall decomposition	Generated (s)	Stored (s)
Grid/context setup	4.125	40.600
Spool reads	0.368	0.356
Coefficient rotation/layout	22.185	22.185
Inverse GSH/FFT	168.187	103.137
Pointwise contractions	28.677	28.689
Frequency accumulation	2.781	2.716
Output writes	2.865	2.938
Worker residual	1.312	1.292
Worker scheduling/wall difference	0.005	0.007
Consumer preflight/cleanup/other residual	4.518	4.725
Total consumer	235.023	206.645
Process outside consumer	274.210	277.023
Process wall	509.233	483.668

The manual consumer timings are a separate timing layer from MATHPROFILE and must not be added to its already-inclusive consumer coverage. Residuals are explicit: roughly 4.5–4.7 s outside setup/worker intervals and 1.3 s inside each worker remain unclassified. MATHPROFILE root exclusive sums close, but that does not imply every manual sub-operation is individually instrumented. The GSH backend here is the library's Loop plus FFT implementation, not a new external-BLAS GEMM; the native tensor route uses Eigen internal multiplication.

The consumer accumulates matching-frequency products side-by-side and writes each completed side once. It skips the exact-zero guard mixing and ten of twelve structurally zero fluid transforms. It does

not sum the two wavefields independently over frequency before multiplying them: that would change the kernel. Dense cross-degree precontraction instead introduces $O(L^2)$ coupling and loses the low-rank frequency advantage when L greatly exceeds the 52 active bins. Approximate compression would require a new error budget.

5. Validation, unresolved discrepancies and resolution

All 17 final-native jobs and 11 native comparison reports pass; all 19 global jobs and nine internal exact comparisons pass; the post-analysis queue's four additional paired exact checks pass. Native and global scientific-array comparisons are byte-exact where reported, work counters are unchanged by the final allocation fix, recorded inputs remain unchanged and process cleanup is verified. At degree 1536 the energy is $8.5734206967 \times 10^{-10}$ m^{2}s and the $q \cdot p$ versus $2E$ relative residual is 6.03×10^{-16}.

Global scientific promotion remains held. The original-direct versus global comparison at degrees 3 and 12 exceeds the frozen 2×10^{-13} threshold in the L and fluid-bulk channels (roughly 3.1–4.3×10^{-13} scaled differences). Internal threading/storage parity does not clear this separate gate, and no tolerance was relaxed. These differences are distinct from the inherited DSpec/YSpec discrepancies: at 1536 the saved solid-channel scaled differences include about 42.8% in C, 34.1% in F and 15.6% in L on the diagnostic points. The fixed-degree waveform agreement is much better (relative $L^2 \approx 4.64 \times 10^{-5}$, energy difference $\approx 2.68 \times 10^{-5}$). Neither proves local-kernel agreement.[2]

The bounded resolution comparison uses a 720-point source-to-receiver great-circle arc, a 5-degree angular exclusion at each endpoint, and approximate bilinear angular interpolation on two native sides. There is no radial interpolation. Between degrees 128 and 384, max-scaled differences are 8.22% for the selected fluid bulk trace and 8.29–17.65% across active solid channels; norm-scaled differences are 8.66% and 8.93–16.63%. This does not establish convergence. The earlier degree-384 to 1536 energy difference is about 2.93% relative to the degree-1536 value (2.84% relative to degree 384), so those degrees must remain labelled exploratory.

6. Decision and next approved boundary

Use the validated native route for initial exploratory sections and depth slices, retaining raw values, one-sided interfaces, contact omissions and explicit scales. At low degree v4 is currently faster; v5 has the demonstrated large-degree gain. The global route remains experimental until its original-route numerical gate is resolved. Its complete 179-side, degree-1536 output would occupy approximately 151.47 GiB; that requested high-degree full volume has not been materialised in this phase.

The proposed figure configuration keeps the same source, receiver, window and band, with kernel-bandwidth-derived section sampling ($4L + 1$ points around the full great circle) and selected native depths near 96.446, 600 and 1019.368 km. Degree 384/1536 panels must be exploratory. Receiver lifts, Sobolev/Riesz gradients and receiver-point peak convergence remain deferred, as requested; they are not prerequisites for the initial figures.

After figures, the largest remaining native targets are genuine batched SEM recovery, angular synthesis and numerical LU. A vertical-receiver adjoint specialisation is a plausible further reduction, but has not been implemented or validated here. The low-degree assembly/scaling/LU regression needs a bounded compiler/memory investigation if further acceleration is requested. No additional optimisation campaign or figure generation starts automatically.

[2] The high-degree native samples occupy only the first 64 angular points of the relevant rule; these discrepancies are descriptive, not a representative global error norm. Native/interface identities are matched without radial interpolation.

A. Complete degree-1536 native exclusive component table

All entries below are exclusive seconds; zero denotes no entry under that scope name, not proof that corresponding work costs zero. Historical scope names are retained. Calls are for v5. Inclusive timings and all other measured degrees remain in the accompanying machine-readable files.

Scope	Original	v4	v5	Calls
SEM_mesh_and_matrix_setup	0.003	0.003	0.003	1
baseline_forward_receiver_only	1.07e-04	0	0	–
baseline_forward_with_bounded_response_capture	0	1.58e-04	1.32e-04	1
batch_response_cache_get_or_compute	0	3.362	3.518	322560
dspec_setup	1.21e-05	1.68e-05	1.12e-05	1
dynamic_H_assembly	31.828	16.116	16.276	162921
forward_frequency_bin	0.702	0.705	0.751	53
forward_substitution	0.118	0.050	0.050	162921
frequency_workspace_modal_solve_and_adjoint_construction	1.29e-04	1.10e-04	1.43e-04	52
frequency_workspace_prepare_and_modal_capture	11.231	10.001	11.006	105
full_weighted_HH_solve	3.02e-05	4.41e-05	5.44e-05	52
harmonic_synthesis_native_node_sampling	1.59e-04	8.91e-05	1.19e-04	52
input_parameters_model_and_frequency_initialization	0.002	0.002	0.002	1
kernel_frequency_sum_read	0.019	0.021	0.021	52
kernel_frequency_sum_write	0.010	0.010	0.011	52
kernel_grid_and_radial_metadata_write	0.003	0.004	0.003	1
kernel_result_file_open_and_zero_initialization	7.08e-04	6.36e-04	0.001	1
measurement_filter_and_inverse_fft	0.002	0.002	0.003	1
minimal_unit_basis_HH_solve	0.062	0.049	0.046	162816
modal_projection_source_lift_and_frequency_solve	0.356	0.317	0.213	105
native_radial_trace_metadata_write	6.38e-04	5.91e-04	6.29e-04	1
packed_Wigner_d_table_build	0.023	0.033	0.033	108
packed_angular_coefficients_and_basis_build	0.010	0.014	0.013	108
parameter_file_generation	2.11e-04	2.17e-04	2.21e-04	1
physical_case_total	0.638	0.933	1.138	1
physical_energy_adjoint_all_output_bins	0.002	0.002	0.003	1
physical_result_and_spectrum_csv	1.76e-04	6.31e-04	0.003	1
pointwise_strain_rotation_kernel_contraction	0.009	0.010	0.010	52
pole_frame_sampling_total	0.010	0.012	0.012	105
radial_interpolation_and_strain_numerator_build	64.612	64.962	66.655	18722
radial_l0_sampling_and_interpolation	0.012	0.019	0.015	105
radial_l0_source_and_adjoint_branch	0.004	0.056	0.074	105
receiver_at_pole_closed_form	0.012	0.025	0.025	322560
receiver_basis_and_QR_reconstruction	5.581	11.880	11.706	325632
receiver_basis_mix_and_unit_adjoint_projection	1.078	3.310	3.406	322560
receiver_frame_geometry_phase_spin_rotation_and_scatter	4.088	4.000	4.050	105
receiver_local_weight_materialization	0.012	0.024	0.025	322560
shared_forward_and_adjoint_solver	2.632	1.287	1.341	162921
solver_factorization_dispatch	0.059	0.031	0.031	162921
source_frame_geometry_phase_spin_rotation_and_scatter	7.977	6.424	7.821	105
sparse_adjoint_solve_including_rhs_scaling	8.955	9.196	9.071	162868
sparse_equilibration_and_scaling	23.019	11.223	11.769	162921
sparse_factorization_total	0.469	0.384	0.376	162921
sparse_forward_solve_including_rhs_scaling	27.815	14.158	14.122	162921
sparse_numerical_factorization	98.255	50.591	52.777	162921
sparse_symbolic_analysis	23.257	2.79e-04	2.88e-04	3
spheroidal_source_lift_projection_and_solver	12.757	75.805	21.302	161280
tensor_angular_gemm_complex_and_packing	64.685	63.747	63.930	91740
tensor_angular_plan_key_build_and_lookup	4.09e-04	4.40e-04	4.90e-04	104
tensor_coefficient_layout_dedup_and_permutation_build	0	0.548	0.548	104
tensor_coefficient_packing_and_key_reorder	134.740	2.607	2.716	624
tensor_cold_frame_geometry_and_wigner_build	0.772	0.696	0.734	53
tensor_complex_basis_cache_lookup	0.003	0.003	0.003	624
tensor_complex_basis_cold_fill_and_cache	4.421	4.316	4.474	521
tensor_complex_basis_key_construction	0.019	0.018	0.019	624
tensor_grid_cold_layout_map_build	0.001	0.003	0.004	1
tensor_grid_semantic_key_build_and_lookup	0.010	0.009	0.009	52
tensor_phase_spin_rotation_and_scatter	0.093	0.266	0.358	104
tensor_radial_l0_scatter	0.031	0.050	0.069	52
tensor_representative_and_coefficient_record_capture	8.74e-04	0.001	0.001	52
tensor_result_allocation_and_volume_tagging	0.025	0.013	0.013	52
tensor_sampling_total	0.331	0.028	0.396	52
toroidal_source_lift_projection_and_solver	1.984	12.744	2.718	161280
trace_csv_write	0.001	0.002	0.003	1
windowed_energy_evaluation	4.91e-07	6.21e-07	6.31e-07	1

B. Complete degree-1536 YSpec kernel components

Scope	Exclusive (s)	Calls
angular.generalized_legendre_table_population	0.787	104
angular.spin_legendre_table_population	0.426	104
angular.wigner_function_table_build	1.591	104
api.ComputeSampledKernels	2.09e-05	1
api.ComputeSampledKernelsImpl	0.382	1
io.output_angle_block_write	1.77e-04	1
io.output_fsync	0.005	1
io.scratch_block_read	0.022	52
io.scratch_block_read_for_transform	2.31e-04	1
io.scratch_block_write	0.014	52
io.scratch_cleanup_after_verified_output	1.64e-04	1
io.scratch_fsync	0.005	1
kernel.angular_block_batch	0.345	52
kernel.angular_harmonic_synthesis.adjoint	22.449	595712
kernel.angular_harmonic_synthesis.source	37.409	595712
kernel.elastic_energy_pair_contraction	0.081	595712
kernel.frequency_block_accumulation_copy	0.002	52
kernel.measurement_frequency_basis_weights	1.76e-05	52
kernel.modulus_dispersion_frequency_weights	3.02e-04	52
kernel.objective_weight_and_parameter_products	0.099	595712
kernel.output_block_parameter_transform	9.75e-04	1
kernel.output_streamed_parameter_transform	3.84e-04	1
kernel.relative_spin_rotation	0.092	1191424
kernel.scratch_block_accumulation	0.009	52
physical_case_total	2.32e-04	1
radial.bvp_up_down_integration_and_matching	1.598	159796
radial.coefficient_combine_and_strain_recovery	11.706	79924
radial.context_material_frequency	0.012	79924
radial.determinant_jumps_matching_and_response_transport	12.050	159796
radial.initial_conditions_and_downward_integration	57.929	159796
radial.initial_conditions_and_upward_integration	1005.270	159796
radial.radial_physical_variable_recovery	0.001	52
radial.radial_response_family	1.36e-04	52
radial.solve_radial_family	5.41e-05	52
radial.solve_toroidal_and_spheroidal_families	0.037	79872
radial.spheroidal_b_vector_transport	2195.872	7468046
radial.spheroidal_physical_variable_recovery	5.604	79872
radial.spheroidal_response_family	0.802	79872
radial.toroidal_physical_variable_recovery	1.355	79872
radial.toroidal_response_family	0.085	79872
setup.construct_kernel_solver	4.17e-05	1
setup.disk_backed_accumulator	3.27e-04	1
setup.frequency_sampling_grid	7.00e-08	1
setup.frozen_measurement_and_sample_grid	0.389	1
setup.map_adjoint_receiver_depths	4.00e-08	1
setup.measurement_adjoint_spectra_and_weights	1.08e-04	1
setup.measurement_direction_basis	3.16e-06	1
setup.merge_sample_nodes	3.40e-05	1
setup.model_calculation_source_receiver_grid	0.001	1
setup.sampled_node_materials_and_phases	1.58e-05	1
setup.source_receiver_frame_geometry	3.44e-05	1
setup.validate_measurements_and_source	3.46e-06	1
source_weights.forward_moment_force_rhs	0.051	79924
source_weights.receiver_unit_force_rhs	0.064	79924

C. Saved evidence and provenance

The report data are derived from saved receipts, not rerun solvers. The independent final review qualifies execution/internal parity as passing and retains the separate global scientific failure. Its local rehash covers the collected small files; large kernel/spool payloads remain on Tunya, bound through verified remote artifact digests and comparison receipts. No production research checkout, existing evidence, commit or remote branch was changed by the acceleration phase.

Companion files: `analyses/phase-a-final-comparison.json` contains all source datasets and input SHA-256 values; `analyses/phase-a-all-profile-components.csv` retains every inclusive/exclusive profiler row at all measured degrees. `review/phase-a-final-evidence-review.md` states the independent audit limits. Proposed figure settings are in `configs/proposed-figure-case.json`. All paths are relative to the acceleration investigation folder.